

KARANJA JAMES THANJU

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PROFILE

Versatile IT Professional and BSc IT graduate with a deep specialization in Platform Engineering and Full-Stack Development. Experienced in building robust applications (Python/React) and the automated infrastructures (Docker/Prometheus) that run them. Proven track record in maintaining high-availability systems for 300+ users while implementing modern DevOps workflows to bridge the gap between code and operations.

EDUCATION

BSC INFORMATION TECHNOLOGY

KCA UNIVERSITY Jan 2023-Nov 2025

Distinction: Second class honours (upper Division)

Kenya Certificate of Secondary Education (KCSE)
Kalou High School 2015 - 2018

Mean Grade: C+

TECHNICAL SKILLS

Platform Engineering & DevOps

- **Containerization:** Docker, Docker Compose, Container Lifecycle Management, Image Optimization.
- **Orchestration & Management:** Portainer (GUI Management), Docker Networking, Volume Persistence.
- **Observability Stack:** Prometheus (Time-series metrics), Grafana (Custom Dashboards), Elasticsearch (Log Aggregation), Blackbox Exporter (Endpoint Probing).
- **Web Servers & Proxies:** Nginx Proxy Manager, Reverse Proxy Configuration, Load Balancing, Header Management.
- **Secure Ingress:** Tailscale Funnel (Zero-Trust), SSH Tunneling, Port Forwarding Alternatives.

Software Development

- **Languages:** Python (Advanced Scripting & Flask),(OOP), JavaScript/ES6+, Bash/Shell Scripting.
- **Frontend:** React (Basics), HTML5, CSS3 (Responsive Design), Bootstrap/Tailwind.
- **Backend & Data:** RESTful API Design, SQL Database Management (SQLite, PostgreSQL), SQLAlchemy (ORM).
- **Methodologies:** (OOP), MVC Architecture, Software Design Patterns, SDLC.
- **Version Control:** Git, GitHub (Branching, Merging, Pull Requests), Webhooks.

IT Operations & Professional Support

- **Operating Systems:** Linux (Ubuntu/Debian Administration, CLI proficiency), Windows Server 2019/2022, MacOS.
- **Network Engineering:** TCP/IP Protocols, LAN/WAN Configuration, DNS/DHCP Management, Firewall Rules, CGNAT Navigation.
- **Hardware/Support:** Tier 1 & 2 Technical Support, Preventative Maintenance, Hardware Diagnostics, Peripheral Troubleshooting.
- **ITSM & Security:** Incident Management (Ticketing Systems), System Hardening, SSL/TLS Certificate Management (Let's Encrypt), Environment Variable Management (Secret Security)

EXPERIENCE

Potterhouse school and Kindergarten

Industrial attachment-IT helpdesk Attachee Sep2025 - Nov 2025

- **Infrastructure Availability:** Managed Tier 1 technical operations for 300+ users, ensuring high availability of core classroom and network services by resolving critical outages with a focus on Mean Time to Recovery (MTTR).
- **Performance Optimization:** Enhanced system stability by diagnosing TCP/IP network bottlenecks and implementing automated preventative maintenance protocols, reducing recurring system downtime.
- **Systems Troubleshooting:** Applied advanced analytical and system-level diagnostics in high-pressure environments to maintain 99% uptime for digital learning platforms and administrative servers.

PROJECTS

Homelab SRE Orchestrator & Observability Stack 2026

- **Infrastructure as Code (IaC):** Engineered a containerized environment using Docker Compose to manage a suite of 10+ services, ensuring reproducible deployments and isolated environments.
- **Full-Stack Observability:** Implemented a professional monitoring pipeline using Prometheus for metric collection and Grafana for real-time visualization of server health (CPU, RAM, Disk I/O).
- **Log Management:** Configured Elasticsearch for centralized log aggregation, allowing for rapid troubleshooting and system behavior analysis across all running containers.
- **Secure Networking:** Deployed a Double-Reverse Proxy architecture using Nginx Proxy Manager and Tailscale Funnel, enabling secure public access to internal services without risky port forwarding.
- **CI/CD Workflow:** Established a "Live-Sync" development-to-production pipeline using Git and Docker bind-mounts, achieving near-zero deployment time for application updates.

IT Ticketing System (Full-Stack Web Application)

- **Development:** Architected a role-based support platform using Python (Flask) and SQLAlchemy, featuring a secure dashboard for tracking IT requests across a 300-user environment.
- **Database Design:** Optimized data persistence by designing a relational schema in SQL, ensuring data integrity for ticket logging, user authentication, and audit trails.
- **Security & Hardening:** Applied "Shift-Left" security principles, implementing Environment Variables for secret management and hardening the Nginx configuration for production-grade deployment

CCTV-AI sytem-2025

- **Automation:** Built a real-time security monitoring solution using Python and Computer Vision (OpenCV) to detect unauthorized access and trigger instant mobile alerts.
- **System Reliability:** Developed integrated logic to manage high-bandwidth video feeds efficiently, focusing on system uptime and continuous operations planning.

REFEREES

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